

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

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Exam Equivalent Time: 11.25 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-08573

Place	Population
Melbourne	3 879 261
Sandleford	2 321 898
Brownton	2 205 873
Chelsea	1 099 604
Priory	963 740
Hunter	308 997
Silby	444 820
Karuah	22 317

How many **more** people live in Sandleford than Silby and Karuah combined?

467 137 422 503 1 854 761 1 899 395

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2. Number and Algebra, NAP-02654

On a country property, 1 acre of land is recommended for every 4 sheep.

How many acres of land would be needed for 16 sheep?

4 6 16 64

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3. Number and Algebra, NAP-55613

The first three days of the Brisbane cricket test had the following attendances:

Crowd Size	
Day 1	20 156
Day 2	18 397
Day 3	29 981

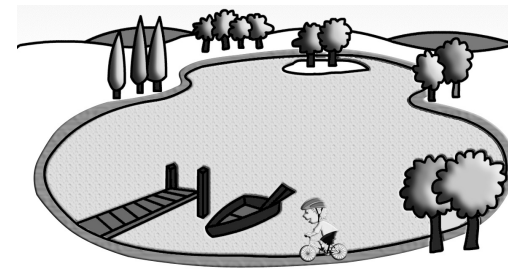
What was the total crowd over the first 3 days, to the nearest 1000?

68 000 69 000 70 000 71 000

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4. Number and Algebra, NAP-02653

It takes Cadel 1.6 minutes to ride his bike one full lap around a lake track.



How many laps does Cadel complete in 8 minutes?

2 5 12 13

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5. Number and Algebra, NAP-26178

Which number shows 5 thousands, 9 tens and 3 tenths?

590.03 5090.3 5009.3 5090.03

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6. Number and Algebra, NAP-90995

Here is a table of values for x and y .

x	0	0.5	1	1.5	2
y	0	1	4	9	16

Which of these is a correct rule for y in terms of x ?

- $y = x$ $y = 2x$ $y = 6x$ $y = 4x^2$
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7. Number and Algebra, NAP-91025

A large telescope lens has a diameter D and depth d .

The focal length of the lens, F is given by the rule $F = \frac{D^2}{9d}$.

When $D = 6$ and $F = 0.4$, the value of d is

- 0.4 0.1 1.7 10
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8. Number and Algebra, NAP-79196

Scientists estimate that in 1 km² of swamp, 30 000 mosquito eggs are laid every minute.

Which value is the closest to the number of mosquito eggs that would be laid in one day?

- 7.2×10^4 7.2×10^5 4.32×10^5 4.32×10^7
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9. Number and Algebra, NAP-61591

The daily energy requirement, E (kilojoules), for a person of mass m (kilograms) is calculated using the rule $E = 7m + 7300$.

For Elijah, $E = 7755$

What is Elijah's mass?

kilograms

Worked Solutions

1. Number and Algebra, NAP-08573

$$\begin{aligned}\text{Number of people} &= 2\,321\,898 - (444\,820 + 22\,317) \\ &= 1\,854\,761\end{aligned}$$

2. Number and Algebra, NAP-02654

$$\begin{aligned}\text{Acres} &= \frac{16}{4} \\ &= 4\end{aligned}$$

3. Number and Algebra, NAP-55613

$$\begin{aligned}20\,156 + 18\,397 + 29\,981 \\ &= 68\,534 \\ &= 69\,000 \text{ (nearest 1000)}\end{aligned}$$

4. Number and Algebra, NAP-02653

$$\begin{aligned}\text{Laps} &= 8 \div 1.6 \\ &= 5 \text{ laps}\end{aligned}$$

5. Number and Algebra, NAP-26178

$$5090.3$$

6. Number and Algebra, NAP-90995

By trial and error for each given equation:

$$\begin{aligned}\text{Consider } y &= 4x^2, \\ 0 &= 4 \times 0^2 = 0 \\ 1 &= 4 \times 0.5^2 = 1 \\ 4 &= 4 \times 1^2 = 4 \text{ etc ...}\end{aligned}$$

$$\therefore y = 4x^2 \text{ is the correct rule.}$$

7. Number and Algebra, NAP-91025

$$0.4 = \frac{6^2}{9 \times d}$$

$$9d = \frac{36}{0.4}$$

$$\begin{aligned}\therefore d &= \frac{90}{9} \\ &= 10\end{aligned}$$

8. Number and Algebra, NAP-79196

Total eggs laid in one day

$$= 30\,000 \times 60 \times 24$$

$$= 4.32 \times 10^7$$

9. Number and Algebra, NAP-61591

$$7755 = 7m + 7300$$

$$7m = 455$$

$$m = \frac{455}{7}$$

$$= 65 \text{ kilograms}$$